

PIETER AYAL

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EDUCATION

University of Manchester

Manchester, United Kingdom

Bachelor of Science (Honours) in Computer Science with Industrial Experience

Sep. 2024 – Jun. 2028

- Academic Performance (First Class): First Year: **75.3%** | Second Year: **76.8%**
- Relevant Coursework: Data Structures and Algorithms, Compilers, System Architecture, Concurrent Programming, Distributed Systems, Operating Systems, Software Engineering

EXPERIENCE

Amazon Web Services

Dublin, Ireland

Software Development Engineering Intern

Jan. 2027 – Jun. 2027

- Incoming internship working on **AWS ElastiCache** in **Java** on the **Control Plane** team.

SLB

Abingdon, United Kingdom

HPC Software Engineering Intern

Sep. 2026 – Dec. 2026

- Incoming internship focused on **upscaling embedded meshes** in **C++** for **HPC simulation** workflows.

Thought Machine

London, United Kingdom

Software Engineering Intern

Jun. 2026 – Sep. 2026

- Architected and developed a full-stack **Balances Visualiser** in **TypeScript** and **React** to accelerate incident debugging for on-call engineers and onboard new engineers onto the ledger's complex accounting model.
- Drove the design end-to-end, authoring an RFC reviewed across teams and landing the implementation through cross-team code and technical reviews.
- Built a **parsing layer** that validates financial transaction payloads against the internal banking engine's **REST API** schema, catching malformed or non-conformant data.

PROJECTS

Distributed Systems Challenges (Fly.io Gossip Glomers) | Go | github.com/pieter124/fly.io-gossip-glomers

- Implemented a fault-tolerant broadcast protocol using a fanout-tree topology (branching factor 4) with per-neighbour batching and retry-until-acknowledged delivery.
- Built a partitioned **Kafka-style commit log** with hash-based key ownership and RPC forwarding across nodes, verified with Maelstrom's Jepsen-based checker across **18,621** operations with **zero failures** and **99.97%** availability (1.24 inter-server msgs/op).

Sharded Concurrent Ledger | Go, gRPC, Protobuf | github.com/pieter124/concurrentbankledger

- Built a **thread-safe** banking engine in **Go**, exposing **gRPC** endpoints for account initialisation and transfers.
- Designed and benchmarked three concurrency strategies: fine-grained per-account mutexes (~1450 ns/op), a single-actor model (~2225 ns/op), and a lock-free sharded-actor architecture (~1080 ns/op).
- Implemented a two-phase reserve/credit/finalise protocol for cross-shard transfers, ensuring zero-sum conservation under concurrent load.
- Engineered a write-ahead log (append + fsync, JSON records) to provide crash recovery, idempotency, and state rebuilding on restart.

EXTRA-CURRICULAR ACTIVITIES

Google Student AI Hackathon 2026 | 3rd Place | github.com/pieter124/google-ai-hackathon-2026

Hackabot (Robotic Arm Track) 2026 | 2nd Place | github.com/pieter124/hackabot26

SAT Solver Competition 2025 | 2nd Place | github.com/pieter124/rustSAT

HPC Summer School 2025 | Edinburgh Parallel Computing Centre, University of Edinburgh

TECHNICAL SKILLS

Languages: Go, C++, TypeScript, Python, Rust, C, Java

Frameworks & APIs: React, gRPC, REST

Parallelism: OpenMP, MPI, SLURM

Tools: Protobuf